FLEET TEAM DIGEST

737MAX-FTD-34-17003

Issue Title : Flight Management Computer System (FMCS) U13 In Service Issues

Airplane Model	ATA	Minor Model(s)
737MAX	3460-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
04/06/2017	01/11/2021	01/11/2021

Estimated Completion Date	Next Update Date

Revision Description

Added FMC OPS U14 service bulletin release information and closed FTD.

Applicability

All 737 MAX airplanes.

Description

The following issues have been reported in service or during simulator/flight test associated with the

U13 FMC software installed on the 737 MAX airplane:

1. Airplane Descent during VNAV after calculated Top of Descent (TOD) point.

2. Takeoff thrust isn't tapering into Maximum Continuous Thrust (MCT) based on altitude or Mach number (only applicable for 737 MAX airplanes with the High Altitude option).

3. MDS and FMCS display unit (MCDU/CDU) display of the "FMC DISAGREE" message.

4: Reported Map Shifts and/or Ground Speed spikes occurring on ground.

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Background

Issue 1: During the 737 MAX flight test program, the flight test crew reported an occasion where the FMC did not descend at its expected TOD point.

The U11 FMC software introduced a change identified as this condition's root cause. This change is also present in the U13 FMC software that is installed as basic on the 737 MAX airplane.

The change incorporated in U11 (which is also present in U12 and U13 FMC software) adds periodic FMC performance re-predictions. Prior to U11, performance predictions were triggered as a result of crew action/flight plan changes.

U11 and later FMC software will periodically re-predict performance to better account for changing environmental conditions in addition to re-predicting performance as a result of crew action/flight plan changes.

For the NG airplane, performance predictions are performed faster and have not resulted in any related in-service reports. However, the time required to perform predictions increased for the 737 MAX airplane. This can result in the FMC performing its path prediction calculations at the same time the airplane is sequencing its TOD point. As a result there have been a few reports during the 737 MAX flight test program where the FMC would initiate predictions just before transitioning into descent. The updated reference path outcome from predictions provides a new path when the airplane is already in its descent phase. When this occurs, it can lead to a 600 to 800 foot VNAV path discrepancy. This condition is apparent at TOD only. Based on the prediction outcome, the airplane can find itself below or above path.

Boeing and GE Aviation have developed a fix for this condition, which will be introduced in the U14 FMC software update (ref: 737MAX-FTD-34-18005).

The mitigation for this condition, may require crew tactical intervention to re-establish path tracking. If below the descent path, the FMC will provide guidance to the DFCS to command a 1000 fpm descent until VNAV path intercept. If above path with COMMON VNAV option enabled, the FMC may find itself in late descent where the guidance mode pitch command is speed on elevator (VNAV SPD as seen on the FMA). Crew intervention may be required to recapture the path.

Issue 2: 737 MAX FMC U13 software incorrectly continues to command takeoff thrust instead of transitioning to climb thrust. The software incorrectly computes the takeoff thrust without the takeoff-to-climb N1 blend at high altitude and high speed

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This issue is only related to 737 MAX airplanes equipped with the High Altitude option and only when performing a takeoff using full rated takeoff thrust at an airport above a pressure altitude of 8,400 ft. The condition is not present in the 737NG airplane.

Issue 3: When in the Approach phase or when on the ground, the FMC DISAGREE message will be displayed to indicate there is a disagreement between the two FMCs. During a disagree event, the FMC will attempt to resync to address the condition. There are a number of conditions that can create an FMC Disagreement. The attachment to this FTD provides several reported FMC DISAGREE events. The attachment includes available mitigation and corrective action planned to address the noted conditions.

Issue 4: 737MAX Operators have reported Map Shifts and/or Ground Speed spikes and Speedbrake warnings occurring on ground. The root cause for these reports is associated with the Auto Navigation Realignment (ANR) function unique to the Integrated ADIRU (I-ADIRU), P/N HG2050BC04. After landing or during taxi out when the airplane is stationary for 7.5 minutes or longer, the I-ADIRU can perform an automatic alignment based upon GPS position data available to the unit. This can result in FMC MAP shifts and/or residual ground speed spikes shortly after landing or during taxi when the airplane position remains static.

Status

Issues 1 and 2 will be addressed in the next FMC software update, U14, which is applicable to NG and MAX airplanes. Details for U14 can be found in Boeing 737 MAX FTD, 737MAX-FTD-34-18005.

Issue 3 as discussed in the attachment to this FTD, there are several conditions that can result in display of FMC DISAGREE. The attachment provides details related to reported FMC DISAGREE messages, available mitigation and planned fixes.

Issue 4 will be addressed in the U14 FMC software update. Boeing has released Flight Operations Technical Bulletin, 18-01, dated 19 March 2018 titled, Integrated ADIRU (I-ADIRU) and Automatic Navigation Realign (ANR) Mode. This document discusses this condition and associated flight deck effects.

Interim Action

No interim action is required for Issue 1. As previously mentioned, mitigation may require crew tactical

intervention to re-establish path tracking. If below the descent path, the FMC will provide guidance to the

DFCS to command a 1000 fpm descent until VNAV path intercept. If above path with COMMON VNAV option

enabled, the FMC may find itself in late descent where the guidance mode pitch command is speed on elevator

(VNAV SPD as seen on the FMA). Crew intervention may be required to recapture the path.

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For Issue 2, Boeing plans to update the 737 MAX AFM on airplanes equipped with the High Altitude Option with the following performance condition note:

Performance Conditions and Procedures - Climb Out

CONDITIONS

Climb gradient and obstacle clearance flight path performance are based on failure of the most critical engine at VEF (Assumed Critical Engine Failure Takeoff Speed).

When performing a takeoff using full rated takeoff thrust at an airport above a pressure altitude of 8,400 ft, ensure that the portion of the Takeoff Flight Path data based on the use of takeoff thrust is completed by an airplane pressure altitude of 19,000 ft. This can be verified by ensuring that the AFM-DPI flight path points labeled as “Level-Off” and “Start Final Climb” (if present) do not exceed an airplane pressure altitude of 19,000 ft. Use of fixed derate or assumed temperature thrust reduction will result in AFM-DPI performing this check.

Issue 3: Boeing investigated the reports of FMC DISAGREE messages that are occurring in-service as discussed in this FTD's attachment.

Issue 4: Operators should review Boeing Flight Technical Bulletin, 18-01 dated 19 March 2018 related to the IADIRU ANR function and associated flight deck effects. This condition is only applicable to 737MAX airplanes equipped with I-ADIRU P/N HG2050BC04 and is not applicable to the 737NG airplane.

Final Action

Issue 1, 2 and 4 are addressed in FMC software update, U14 (FTD 737MAX-FTD-34-18005).

Details related to Issue 3 are provided in the attachment to this FTD.

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Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

U14 Service Bulletin Available: 9 December 2020 (Complete)

U14 Parts Available: 9 December 2020 (Complete)

Operator Action

When operators report display of the FMC DISAGREE message during flight or on the ground, FMC Fault Log data can be reviewed to assist in determining the disagreement's root cause. Operators can provide FMC Fault Log data, per AMM TASK 34-61-00-810-801 or FMC Diagnostic Data Transfer via Network File Server, TASK 34-61-00-810-802. The downloaded files can be provided to Boeing and/or to GE Aviation. Files can be submitted direct to GE Aviation using the email address:

737fmchelp@ge.com

Boeing recommends downloading the data as soon as practical after the event within the next available maintenance opportunity. However, the data typically is retained in FMC memory for one month (possibly longer) after an occurrence.

Operators should review Boeing Flight Technical Bulletin, 18-01 dated 19 March 2018 related to the I-ADIRU ANR function and associated flight deck effects.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-34-16010		
Flight Operations Technical Bulletin (FOTB)	18-01		03/19/2018
Fleet Team Digest Article (FTD)	737MAX-FTD-34-18005		
Service Bulletin (SB)	737-34-3065		12/09/2020

Related Categories

- In Service Occurrences : null

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Attachments:

a) 737MAX FMC Disagree Message

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